Enabling Scalable Distributed Beamforming via Networked LEO Satellites Toward 6G

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***Abstract*—In this paper, we propose scalable distributed beamforming schemes over networked low Earth orbit (LEO) satellite systems that rely solely on statistical channel state information (CSI). We begin by introducing the LEO satellite network system model and presenting pragmatic yet effective analog beamformer and user-scheduling designs. We then derive a closed-form lower bound on the ergodic sum rate, based on the hardening bound, using which we formulate a per-satellite power-constrained sum rate maximization problem for the digital beamformer design. Next, we provide a centralized solution, obtained via the weighted minimum mean squared error (WMMSE) framework, establishes performance limits and motivates decentralized strategies. We subsequently introduce two decentralized optimization schemes, based on approximating the hardening bound and decentralizing the WMMSE framework, for two representative inter-satellite link (ISL) topologies, i.e., Ring and Star topologies. In the Ring topology-based beamforming scheme, satellites update beamformers locally and exchange intermediate parameters sequentially. On the other hand, in the Star topology-based beamforming scheme, edge satellites update beamformers locally and in parallel, achieving consensus on intermediate parameters at a central satellite using a penalty-dual decomposition (PDD) framework. Extensive simulations demonstrate that the proposed distributed beamforming schemes achieve similar performance with the centralized beamforming scheme while improving scalability significantly. Additionally, we reveal the delay–overhead trade-off between the two topologies.**

***Index Terms*—LEO satellite communication, distributed beamforming, ISL, multi-satellite networks.**

I. Introduction

5G IS being deployed at an unprecedented pace, transforming daily life and driving innovation across global vertical industries. However, a significant digital divide persists. According to data released by the International Telecommunication Union (ITU) in 2023, approximately one-third of the global population, about 2.6 billion people, remain offline [1]. Bridging this gap remains a core ambition for 6G, with the vision to *connect the unconnected* [2]. Early 6G initiatives, both academic and industrial [3], [4], have already emerged, with ubiquitous connectivity identified as one of the six key dimensions of the "6G wheel" defined by the ITU. From a standardization standpoint, the 3rd Generation Partnership Project (3GPP) has launched a study item on 5G non-terrestrial networks (NTN), aiming to integrate satellite systems into mobile broadband and machine-type communication scenarios [5]. It is widely anticipated that low Earth orbit (LEO) satellite networks, as a fundamental component of NTN, will be essential to achieving ubiquitous coverage in 6G and beyond, a feat unachievable by terrestrial networks alone.

LEO satellite systems feature dense constellation deployment, enhanced signal strength, and lower latency due to their reduced orbital altitude. As a result, they have already surpassed traditional medium Earth orbit (MEO) and geostationary Earth orbit (GEO) satellite systems in terms of communication quality. However, the conventional paradigm, where each user terminal (UT) is served by a single LEO satellite, faces inherent limitations in power budget and array aperture. This poses challenges in meeting the growing demand for high-throughput communications using a single LEO satellite [4]. Consequently, there is increasing interest in leveraging the cooperative capabilities of multiple LEO satellites, interconnected via inter-satellite links (ISLs), to jointly serve UTs [6], [7], [8], [9], [10], [11], [12], [13], [14], [15]. For examples, the authors of [6] and [7] introduce the distributed massive multi-input-multi-output (MIMO) concept, which allows UTs to be served by networked LEO satellites. Their results show that cooperation forms a virtually enlarged antenna array, thereby enabling distributed beamforming and enhancing communication rates. In [8], the impact of the geometrical formation of these distributed arrays, hosted by networked satellites, on downlink communication throughput is thoroughly analyzed. In [9] and [10], the authors exploit the antenna arrays at each UT to spatially filter signals from multiple LEO satellites and to align them in both time and frequency domains. The optimality of single-stream transmission from each satellite to each UT for maximizing the ergodic sum rate of networked LEO satellite distributed beamforming is validated in [9], while [10] devises a hardware-friendly analog beamforming scheme. The works in [11] and [12] leverage the line-of-sight (LOS)-dominant nature of the LEO satellite channel and use UT position information to facilitate efficient distributed beamforming across networked LEO satellites. In [13], a joint hybrid beamforming and user-scheduling scheme is designed for cooperative satellite networks.

Despite recent advancements in networked LEO satellite beamforming, two major limitations persist in conventional

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approaches. First, many existing works rely on instantaneous channel state information (CSI) for beamforming [6], [13], [16], which may be difficult to obtain due to the latency incurred by the channel estimation process and the short coherence time resulting from the rapid movement of LEO satellites [17], [18], [19], [20]. Second, many approaches adopt a centralized optimization paradigm, where each satellite uploads its CSI to a central processing unit (CPU), typically implemented on a central satellite, that jointly computes the beamformers for all satellites and then distributes them back to the corresponding nodes [10], [13], [21]. Given the limited on-board processing capabilities of LEO satellites, this paradigm poses significant scalability challenges, as the central satellite must handle the majority of the computational burden. To address these limitations, several distributed beamforming schemes have been proposed. For simplicity, they typically employ simple linear beamformers, such as maximum ratio transmission (MRT), zero-forcing (ZF), or their variants [6], [7], [8], [13], [14], [21]. While computationally attractive, these heuristic methods may lead to noticeable performance degradation compared to optimization-based methods that more effectively refine the solution toward the desired utility.

Therefore, a critical research gap exists in designing distributed beamforming schemes tailored for LEO satellite networks that do not rely on instantaneous CSI, yet deliver high performance and are scalable for practical deployment.[1] To address this gap, this paper investigates distributed downlink beamforming schemes for scalable networked LEO satellites, aiming to unleash the power of large-scale cooperation over LEO satellite networks. Our goal is not merely to *connect the unconnected*, but to *uplift the connected* with transformative communication quality enhancements in the sky as we advance toward 6G. The main contributions of this paper are summarized as follows.

- We develop a comprehensive system model for networked LEO satellite distributed beamforming, where orthogonal frequency division multiplexing (OFDM) is employed, and each satellite is equipped with a large antenna array and a limited number of radio-frequency chains (RFCs). The model encompasses a thorough analysis of channel characteristics and pragmatic designs for user scheduling and analog beamforming for networked LEO satellite system. To sidestep the need for instantaneous CSI, we resort to the ergodic rate, which is a function of channel statistics, and then derive its closed-form lower bound by invoking the hardening bound. Using the derived ergodic rate expression, we formulate a downlink sum rate maximization problem that optimizes digital beamformers under the power budget constraints of each satellite. We first solve this problem using the weighted minimal mean squared error (WMMSE) framework in a centralized manner, which not only lays the foundation for developing distributed optimization schemes but also establishes an upper bound for their performance;
- We develop scalable distributed optimization approaches tailored to networked LEO satellite beamforming by approximating the hardening bound and modifying the WMMSE framework to enable decentralized processing.[2] In the proposed design, beamformers are optimized locally at the corresponding satellites, while the exchange of intermediate parameters, crucial for achieving coherent utility, follows the information flow dictated by different yet representative ISL topologies, namely the Ring and Star topologies. In the Ring topology, information flows unidirectionally through the network in a sequential manner, which is fully decentralized but may incur longer delays due to propagation time. In the Star topology, bidirectional information exchange occurs between edge satellites and a central satellite node, enabling parallel processing at the edges. This Star topology, however, requires a central node to reach consensus on the intermediate parameters updated simultaneously by the edge nodes, a challenge that is addressed using a penalty dual decomposition (PDD)-based consensus algorithm.
- We conduct extensive simulations on realistic networked LEO satellite systems to validate the effectiveness of the proposed approaches: 1) comparable performance is achieved relative to the centralized counterpart, while preserving scalability; 2) significant performance gains are achieved over conventional LEO satellite distributed beamforming schemes that use heuristic local beamformers, as well as schemes where each UT is served by a single LEO satellite, either with optimized or heuristic beamforming; and 3) a tradeoff between delay and communication overhead is observed across different topologies, offering valuable insights for future research on large-scale cooperation in more complex and dynamic LEO satellite networks.

This paper is organized as follows. We begin by presenting the system model in Section II. Section III proposes a centralized optimization scheme for networked LEO satellite distributed beamforming that maximizes the downlink sum rate. Sections IV and V introduce distributed optimization schemes, based on the Ring and Star topologies of ISLs, respectively, to address the scalability issues of the centralized counterpart. Numerical results are presented in Section VI, and Section VII concludes the paper.

Notations: Scalars are written in regular lowercase, while vectors and matrices are denoted by bold lowercase and bold uppercase letters, respectively. For a vector $\mathbf{a}$, its 2-norm is expressed as $\|\mathbf{a}\|$. For a matrix $\mathbf{A}$, its Frobenius norm is expressed as $\|\mathbf{A}\|_{\mathrm{F}}$. We use $(\cdot)^*$, $(\cdot)^{\mathsf{T}}$ and $(\cdot)^{\mathsf{H}}$ as superscripts to indicate the conjugate, transpose, and Hermitian transpose operations, respectively. $\Re\{a\}$ denotes the real part of a

[1]While our focus is on algorithmic scalability under statistical CSI, practical transceiver/aperture hardware impairments (e.g., phase noise, nonlinearity, mutual coupling) can influence large-array NTN/LEO beamforming [22], [23], [24], [25]. Incorporating such effects into the proposed framework is an important extension left for future work.

[2]Note that distributed implementations of WMMSE have been studied in prior work, e.g., [26]. However, those approaches rely on UT-side covariance estimation and iterative feedback to the transmitters, which is infeasible in LEO satellite systems due to long propagation distances, limited feedback capacity, and excessive latency. In contrast, our design leverages ISLs to achieve decentralized coordination without UT-side involvement.

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TABLE I
LIST OF ABBREVIATIONS

Abbreviation	Full Form
AOD	Angle Of Departure
ALP	Augmented Lagrangian Problem
CPU	Central Processing Unit
CSI	Channel State Information
DOF	Degree Of Freedom
GEO	Geostationary Earth Orbit
GNSS	Global Navigation Satellite System
LEO	Low Earth Orbit
LOS	Line-Of-Sight
MEO	Medium Earth Orbit
MIMO	Multiple-Input-Multiple-Output
MRT	Maximum Ratio Transmission
NLOS	Non-Line-Of-Sight
NTN	Non-Terrestrial Networks
OFDM	Orthogonal Frequency Division Multiplexing
PDD	Penalty Dual Decomposition
QCQP	Quadratically Constrained Quadratic Program
RFC	Radio Frequency Chain
UPA	Uniform Planar Array
UT	User Terminal
WMMSE	Weighted Minimal Mean Squared Error
ZF	Zero-Forcing

TABLE II
LIST OF ESSENTIAL SYMBOLS

Notation	Definition
S	LEO satellite number
U	UT number
$N = N_{\mathrm{h}} \times N_{\mathrm{v}}$	Antenna number (horizontal/vertical)
N_{RF}	RFC number (per satellite)
$T = \min(U, N_{\mathrm{RF}})$	Served UT number (per satellite)
B	System (operating) bandwidth
Δf	Subcarrier spacing
K	Subcarrier number
T_{sym}	OFDM symbol duration
$\boldsymbol{\theta}_{s,u} = [\theta^{\mathrm{az}}_{s,u}, \theta^{\mathrm{el}}_{s,u}]^{\mathsf{T}}$	AOD (azimuth/elevation)
$G(\theta^{\mathrm{el}}_{s,u})$	Antenna radiation gain
$\mathbf{a}(\boldsymbol{\theta}_{s,u})$	Array steering vector
$\alpha_{s,u}[k]$	Composite channel gain (Rician)
$\bar{\alpha}_{s,u}[k]$	Mean of $\alpha_{s,u}[k]$
$\beta_{s,u}[k]$	Variance of $\alpha_{s,u}[k]$
$\mathbf{F}_s = [\mathbf{f}_{s,1}, \ldots, \mathbf{f}_{s,N_{\mathrm{RF}}}]$	Analog beamformer
$\mathbf{W}_s[k] = [\mathbf{w}_{s,1}[k], \ldots, \mathbf{w}_{s,U}[k]]$	Digital beamformer (per subcarrier)
$\boldsymbol{\delta}_s = [\delta_{s,1}, \ldots, \delta_{s,U}]^{\mathsf{T}}$	Scheduler vector

complex number a. The Kronecker and Hadamard products are denoted by $\otimes$ and $\odot$, respectively. $\mathbb{E}[\cdot]$, $\mathbb{V}[\cdot]$, and $\mathrm{vec}(\cdot)$ denote the expectation, variance, and vectorization operators, respectively. For statistical distributions, $\mathcal{CN}(\boldsymbol{\mu}, \mathbf{C})$ represents a circularly symmetric complex Gaussian distribution characterized by mean $\boldsymbol{\mu}$ and covariance matrix $\mathbf{C}$. $\mathbf{1}_N$ and $\mathbf{0}_N$ denote the all-one and all-zero N-dimensional column vectors, respectively. The notation $\mathbf{a} \geq \mathbf{b}$ denotes the inequality between each corresponding element of $\mathbf{a}$ and $\mathbf{b}$.

Tables I and II detail the main abbreviations and essential symbols used throughout this paper, respectively.

II. SYSTEM MODEL

As illustrated in Fig. 1, we consider a networked-LEO satellite system where S LEO satellites serve U UTs in the downlink. Each UT is equipped with a single antenna. Each satellite is equipped with a uniform planar array (UPA) comprising $N = N_{\mathrm{h}} \times N_{\mathrm{v}}$ half-wavelength-spaced antennas, where N_{h} and N_{v} denote the number of antennas in the horizontal and vertical dimensions, respectively. In addition, each satellite is equipped with $N_{\mathrm{RF}} \leq N$ RFCs. In our work, we assume an identical number of antennas and RFCs per satellite for simplicity. Nonetheless, it should be noted that extending the model to scenarios with varying antenna and/or RFC configurations is straightforward. Each satellite serves $T = \min(U, N_{\mathrm{RF}})$ UTs, within overlapping time-frequency resources according to a certain scheduling policy. Meanwhile, by exploiting ISLs, information concerning different UTs is exchanged within the network, enabling each UT to be served by multiple satellites simultaneously.

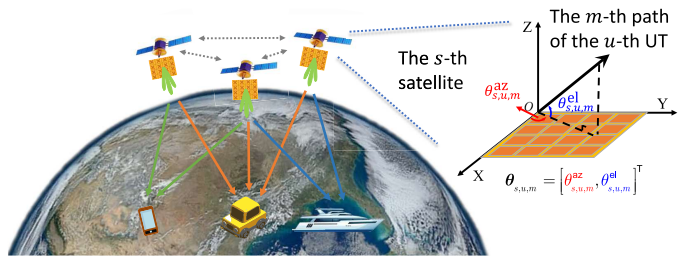

Fig. 1. An illustration of networked-LEO satellite system, where multiple LEO satellites cooperatively serve UTs through distributed beamforming. The geometric configuration of the local coordinate systems for LEO satellites is depicted on the right part, where the UPA is deployed in the XY-plane.

A. Channel Model

1) Downlink Satellite Channel: Let B denote the operating bandwidth of the satellite system. We consider an OFDM system with K subcarriers, and carrier frequency being f_c, where $\Delta f = B/K$ and $T_{\mathrm{sym}} = 1/\Delta f$ represent the subcarrier spacing and symbol duration, respectively. The downlink channel from the s-th satellite to the u-th UT during the ℓ-th OFDM symbol on the k-th subcarrier is modeled as

$$\mathbf{h}_{s,u}[\ell, k] = \sum_{m=0}^{M_{s,u}} \alpha_{s,u,m} G\left(\theta^{\mathrm{el}}_{s,u,m}\right) e^{\jmath 2\pi(\ell T_{\mathrm{sym}} \upsilon_{s,u,m} - k\Delta f \tau_{s,u,m})} \times \mathbf{a}\left(\boldsymbol{\theta}_{s,u,m}\right), \tag{1}$$

where $M_{s,u}$ denotes the number of propagation paths, $\alpha_{s,u,m}$ is the complex channel gain. Here, the LOS path is indexed by $m = 0$, while the remaining paths are non-line-of-sight (NLOS). The parameters $\tau_{s,u,m}$ and $\upsilon_{s,u,m}$ represent the propagation delay and Doppler shift, respectively, and $\mathbf{a}(\boldsymbol{\theta}_{s,u,m}) \in \mathbb{C}^N$ is the steering vector at the satellite, where $\boldsymbol{\theta}_{s,u,m} = [\theta^{\mathrm{az}}_{s,u,m}, \theta^{\mathrm{el}}_{s,u,m}]^{\mathsf{T}}$ denotes the angle-of-departure (AOD), comprising both azimuth and elevation angles. Additionally, $G(\theta^{\mathrm{el}}_{s,u,m})$ denotes the antenna radiation pattern, which depends solely on the elevation angle [27]. Without loss of generality, we assume that the UPA at each satellite is deployed on the XY-plane of its local coordinate system, as illustrated in Fig. 1. Let $\mathbf{n}(N) = [0, \ldots, N-1]^{\mathsf{T}}$. The steering vector is given by

$$\mathbf{a}\left(\boldsymbol{\theta}_{s,u,m}\right) = e^{-\jmath 2\pi \phi^{\mathrm{h}}_{s,u,m} \mathbf{n}(N_{\mathrm{h}})} \otimes e^{-\jmath 2\pi \phi^{\mathrm{v}}_{s,u,m} \mathbf{n}(N_{\mathrm{v}})}, \tag{2}$$

where $\phi^{\mathrm{h}}_{s,u,m} = d \cos\theta^{\mathrm{az}}_{s,u,m} \cos\theta^{\mathrm{el}}_{s,u,m}/\lambda$ and $\phi^{\mathrm{v}}_{s,u,m} = d \sin\theta^{\mathrm{az}}_{s,u,m} \cos\theta^{\mathrm{el}}_{s,u,m}/\lambda$. Here, d denotes the antenna spacing along each dimension, and λ is the wavelength corresponding to the central carrier frequency f_c.

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2) Equivalent LOS Channel: In LEO satellite systems, since the satellite altitude is significantly greater than the distances between the scatterers (typically located on the ground and around the receiver) and the UT, the AODs in (1) can be approximated as identical across all paths, i.e., $\boldsymbol{\theta}_{s,u} = \boldsymbol{\theta}_{s,u,m}, \forall m$. Note also that the Doppler shift associated with each path is induced by the movement of both the satellite and the UT, such that $\upsilon_{s,u,m} = \upsilon^{\text{Sat}}_{s,u,m} + \upsilon^{\text{UT}}_{s,u,m}$, where $\upsilon^{\text{Sat}}_{s,u,m}$ and $\upsilon^{\text{UT}}_{s,u,m}$ denote the contributions to the Doppler shift from the satellite and the UT, respectively. For the same reason as with the AODs, the satellite-induced Doppler shift can be considered identical across all paths, i.e., $\upsilon^{\text{Sat}}_{s,u} = \upsilon^{\text{Sat}}_{s,u,m}, \forall m$. Let $\tau_{s,u} = \tau_{s,u,0}$ denote the minimum propagation delay corresponding to the LOS path, and let $\tau^{\text{Diff}}_{s,u,m} = \tau_{s,u,m} - \tau_{s,u}$ represent the differential delay of the m-th path relative to the LOS path. Then, we can rewrite (1) as the equivalent LOS channel as

$$\mathbf{h}_{s,u}[\ell, k] = \alpha_{s,u}[k]\, e^{\jmath 2\pi\left(\ell T_{\text{sym}} \upsilon^{\text{Sat}}_{s,u} - k\Delta f \tau_{s,u}\right)} \grave{\mathbf{a}}(\boldsymbol{\theta}_{s,u}), \quad (3)$$

where $\grave{\mathbf{a}}(\boldsymbol{\theta}_{s,u}) = G(\theta^{\text{el}}_{s,u})\mathbf{a}(\boldsymbol{\theta}_{s,u})$ and $\alpha_{s,u}[k] = \sum_{m=0}^{M_{s,u}} \alpha_{s,u,m} e^{\jmath 2\pi\left(\ell T_{\text{sym}} \upsilon^{\text{UT}}_{s,u,m} - k\Delta f \tau^{\text{Diff}}_{s,u,m}\right)}$ denotes the composite channel gain, which is modeled as a Rician random variable with Rician factor $\kappa_{s,u}$ and average power $\mathbb{E}[|\alpha_{s,u}[k]|^2] = \gamma_{s,u}[k]$ [10], [28].

It is worth noting that $\alpha_{s,u}[k]$ is affected by the Doppler shift induced by UT movement, and is therefore time-varying. However, since UT movement is moderate in most practical scenarios, we henceforth assume that $\alpha_{s,u}[k]$ is slowly varying and omit its dependence on ℓ. Specifically, the real and imaginary parts of $\alpha_{s,u}[k]$ are independent and identically distributed real-valued Gaussian random variables, each with the mean and variance given by

$$\bar{\alpha}_{s,u}[k] = \sqrt{\frac{\kappa_{s,u}\gamma_{s,u}[k]}{2(1+\kappa_{s,u})}}, \quad \beta_{s,u}[k] = \frac{\gamma_{s,u}[k]}{2(1+\kappa_{s,u})}. \quad (4)$$

The large-scale path loss on the k-th subcarrier, i.e., $\gamma_{s,u}[k]$, can be represented in the dB scale as [5], [10], and [13]

$$-10\log_{10}\gamma_{s,u}[k] = \text{PL}^{\text{FS}}_{s,u} + \text{PL}^{\text{SF}}_{s,u} + \text{PL}^{\text{CL}}_{s,u} + \text{PL}^{\text{AB}}_{s,u} + \text{PL}^{\text{SC}}_{s,u}, \quad (5)$$

where $\text{PL}^{\text{FS}}_{s,u} = 20\log_{10} d_{s,u} + 20\log_{10}(f_c + k\Delta f) - 147.55$ denotes the free-space path loss [5], with $d_{s,u}$ representing the distance from the s-th LEO satellite to the u-th UT. The term $\text{PL}^{\text{SF}}_{s,u}$ captures the shadow fading component, while $\text{PL}^{\text{CL}}_{s,u}$ accounts for the clutter loss. The term $\text{PL}^{\text{AB}}_{s,u}$ models the atmospheric absorption loss, and $\text{PL}^{\text{SC}}_{s,u}$ represents the scintillation-induced attenuation caused by ionospheric and tropospheric irregularities.

Several additional elaborations regarding the above channel model are in order:

- In LOS-dominated scenarios, both shadow fading and clutter loss typically have a negligible impact [29]. Accordingly, we omit these terms from the performance evaluation in our studied scenarios;
- The atmospheric absorption, which depends on signal frequency and satellite elevation angle, is calculated according to ITU recommendations [30] and configured using an off-the-shelf MATLAB script [31];
- Per [5], ionospheric scintillation effects can be ignored for carrier frequencies beyond 6 GHz. As for the tropospheric scintillation term within $\text{PL}^{\text{SC}}_{s,u}$, which lacks a simple analytical form, we adopt the empirical model from [32] as a reference.

Remark 1: From (3), the satellite–UT link includes a phase term due to Doppler shift and propagation delay. In this work, we assume satellite-side, stream-level pre-compensation (timing advance and carrier pre-rotation) toward each UT under a common time/frequency reference (e.g., via global navigation satellite system (GNSS)), so that signals from cooperating satellites arrive delay- and Doppler-aligned at the receiver, avoiding complicated UT-side post-compensation. This idealized compensation is commonly adopted in the LEO-NTN literature (e.g., [9], [10], [11], [28]), where predictable orbital dynamics and UT positioning enable precise correction. Accordingly, (3) simplifies to

$$\mathbf{h}_{s,u}[k] = \alpha_{s,u}[k]\, \grave{\mathbf{a}}(\boldsymbol{\theta}_{s,u}). \quad (6)$$

In practice, small orbital perturbations and oscillator imperfections lead to residual Doppler/delay errors. Within our statistical formulation, these residuals primarily manifest as additional random phase fluctuations in the composite gain $\alpha_{s,u}[k]$. Because our performance metric is the ergodic rate characterized via the hardening bound (introduced later), this metric is typically robust to such instantaneous phase drifts, which slightly alters its second-order statistics. A more detailed modeling and mitigation of imperfect compensation are beyond the scope of this paper and will be investigated in future work.

B. Signal Model

As introduced, the number of RFCs employed by each LEO satellite can be (significantly) smaller than the number of antennas. This is a rational consideration for reducing hardware and energy costs, as each RFC typically incurs high expense and power consumption, which are particularly critical in LEO satellite scenarios due to limited onboard resources and power supply [13]. Consequently, we consider a hybrid beamforming architecture, where the RFCs are connected to the antennas through a cost-effective analog network implemented using phase shifters (PSs), enabling a hardware- and energy-efficient beamforming solution.

Let $\mathbf{s}[\ell, k] = [s_1[\ell, k], \ldots, s_U[\ell, k]]^{\mathsf{T}} \sim \mathcal{CN}(\mathbf{0}_U, \mathbf{I}_U)$ denote the collection of data streams for U UTs during the ℓ-th symbol of the k-th subcarrier. The corresponding transmit signal at the s-th LEO satellite is expressed as

$$\mathbf{x}_s[\ell, k] = \mathbf{F}_s \mathbf{W}_s[k]\, \text{diag}(\boldsymbol{\delta}_s)\, \mathbf{s}[\ell, k], \quad (7)$$

where $\mathbf{F}_s = [\mathbf{f}_{s,1}, \ldots, \mathbf{f}_{s,N_{\text{RF}}}] \in \mathbb{C}^{N \times N_{\text{RF}}}$ denotes the analog beamformer, $\mathbf{W}_s[k] = [\mathbf{w}_{s,1}[k], \ldots, \mathbf{w}_{s,U}[k]] \in \mathbb{C}^{N_{\text{RF}} \times U}$ represents the digital beamformer, and $\boldsymbol{\delta}_s = [\delta_{s,1}, \ldots, \delta_{s,U}]^{\mathsf{T}}$ denotes the scheduler, with each element being either 0 or 1. Specifically, the u-th UT data stream is transmitted via the s-th satellite if $\delta_{s,u} = 1$, and is not transmitted if $\delta_{s,u} = 0$.

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The signal received at the u-th UT during the ℓ-th symbol of the k-th subcarrier is given by

$$\begin{aligned} y_u[\ell,k] &= \sum_{s=1}^{S} \mathbf{h}_{s,u}^{\mathsf{T}}[k]\,\mathbf{F}_s\mathbf{W}_s[k]\,\mathrm{diag}(\boldsymbol{\delta}_s)\,\mathbf{s}[\ell,k] + z_u[\ell,k] \\ &= \sum_{s=1}^{S} \alpha_{s,u}[k]\,\mathbf{g}_{s,u}^{\mathsf{T}}\tilde{\mathbf{w}}_{s,u}[k]\,s_u[\ell,k] \\ &\quad + \underbrace{\sum_{l\neq u}^{U}\sum_{s=1}^{S} \alpha_{s,u}[k]\,\mathbf{g}_{s,u}^{\mathsf{T}}\tilde{\mathbf{w}}_{s,l}[k]\,s_l[\ell,k]}_{\text{Inter-user interference (IUI)}} + z_u[\ell,k], \end{aligned} \tag{8}$$

where $\mathbf{g}_{s,u} = \mathbf{F}_s^{\mathsf{T}}\grave{\mathbf{a}}(\boldsymbol{\theta}_{s,u}) \in \mathbb{C}^{N_{\mathrm{RF}}}$, $\tilde{\mathbf{w}}_{s,u}[k] = \delta_{s,u}\mathbf{w}_{s,u}[k]$, and $z_u[\ell,k] \sim \mathcal{CN}(\mathbf{0},\sigma^2)$ denotes the additive white Gaussian noise (AWGN). Here, the noise variance is given by $\sigma^2 = N_0\Delta f$, where N_0 is the single-sided power spectral density (PSD).

C. User Scheduling

In our work, instead of jointly optimizing the schedulers and beamformers, which typically involves a time-consuming iterative process and incurs significant computational complexity [13], [21], [33], [34], we adopt a more pragmatic scheduling method based on geometric relationships between satellites and UTs. Let $\mathcal{S} = \{1,2,\ldots,S\}$ and $\mathcal{U} = \{1,2,\ldots,U\}$ denote the index sets of LEO satellites and UTs, respectively. For each $s \in \mathcal{S}$, let $\mathcal{U}_s = \{u_s[1],\ldots,u_s[T]\}$ be the index set (in *ascending* order) of UTs served by the s-th satellite. The elements of $\mathcal{U}_s$ corresponds to the T nearest UTs to the s-th satellite, which can be straightforwardly determined by sorting the distances from the UTs to the satellite.

Several points are worth emphasizing: *First*, this scheduling method is independent of CSI and relies solely on *geometric information*, such as the positions of the UTs, which vary slowly (compared to CSI) and can be obtained by the UTs via GNSS and fed back to the LEO satellites[3]; *Second*, beyond its practicality, a key rationale behind this scheduling method is that, in the LOS-dominant satellite communication environment, the link budget is primarily determined by the communication distance. As a result, each satellite can efficiently utilize its resources and enhance the overall system performance; *Third*, when $N_{\mathrm{RF}} \geq U$, i.e., $T = U$, all UTs in $\mathcal{U}$ can be served by each satellite without requiring additional distance-based scheduling. However, since the number of UTs is typically much larger than that of RFCs, this scenario is not particularly relevant in practice.

Remark 2: It should be pointed out that, in dense deployments where multiple UTs are geographically close and thus exhibit nearly identical distances and similar large-scale/channel statistics, nearby UTs are aggregated into a group and the group is treated as a single scheduled entity. A single data stream is then transmitted to the entire group via beam-wise broadcasting (multicast). The distance-based scheduler operates on the group centroids and selects the T nearest groups per satellite, which naturally avoids tie cases among nearly co-located UTs. Crucially, this grouping is agnostic to the subsequent beamforming optimization and a group is indistinguishable from an individual UT after replacing the UT's geometry/statistics with those of the group's centroid (or aggregate statistics). The problem formulation, variables, constraints, and algorithms therefore remain unchanged.

[3]Position feedback may be slightly outdated due to propagation/feedback latency. At LEO ranges (on the order of 500 km), one-way propagation is <2 ms. Even at 20 m/s, a UT moves <0.04 m in this interval, negligible relative to the link distance, so geometry-dependent statistics (and thus statistical CSI) are essentially unaffected. Consequently, modest update rates (sub-second to multi-second, depending on mobility) typically suffice.

D. Analog Beamformer

Inspired by [35] and [36], the analog beamformer is designed using the steering vector as

$$\mathbf{F}_s = \left[\mathbf{a}^*(\boldsymbol{\theta}_{s,u_s[1]}),\ldots,\mathbf{a}^*(\boldsymbol{\theta}_{s,u_s[T]})\right]. \tag{9}$$

Note that $\mathbf{F}_s$ is a function of the AODs $\boldsymbol{\theta}_{s,u}, u \in \mathcal{U}_s$, which are determined by the positions of LEO satellites and UTs. This enables the analog beamformers to be constructed *locally* at each satellite and inherently satisfies the unit-modulus constraint of the PS [37]. In addition, as future LEO satellite systems are envisioned to be equipped with massive antenna arrays [9], [28], [38], [39], this design allows us to harness the large *array gain* provided by the excessive antennas in the analog domain [35], without relying on complex unit-modulus-constrained hybrid beamforming optimization. On the other hand, owing to the LOS-dominant nature of the LEO satellite channel and the asymptotic orthogonality property (i.e., favorable propagation property) [40], the analog beamformer design in (9) enables us to concentrate the power of the equivalent steering vector $\mathbf{g}_{s,u}$ on its u-th element (if $u \in \mathcal{U}_s$), effectively managing IUI in the analog domain. This approach has been shown to achieve performance comparable to even fully digital solutions [36].

III. Centralized Beamforming Optimization and Inter-Satellite Link Topologies

In this section, we first formulate a beamforming optimization problem for networked LEO satellites systems, with the objective of enhancing downlink sum rate. A WMMSE-based solution is then developed, which inherently requires centralized implementation at a CPU, limiting its scalability in large-scale satellite networks. To address this issue, we introduce several distinct yet representative ISL topologies for LEO satellite constellations. These topologies serve as the foundation for developing distributed beamforming schemes based on decentralized optimization, thereby enabling the computational workload to be distributed among all participating LEO satellites.

A. Sum Rate Maximization Problem Formulation

Since acquiring accurate instantaneous CSI is challenging in LEO satellite systems due to short coherence time and long propagation delay (compared to terrestrial counterparts) [17], [18], [19], [20], [39], we opt to optimize a statistical performance metric, typically characterized by the ergodic

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rate, instead of its instantaneous counterpart. Since the ergodic rate usually lacks a closed-form expression, we resort to its lower bound, namely the hardening bound [41], [42], to facilitate tractability.

From (8), the lower bound of the ergodic rate at the u-th UT over the k-th subcarrier is given in (11), shown at the bottom of the page, Here, $\tilde{\mathbf{w}}_u[k] = [\tilde{\mathbf{w}}_{1,u}^{\mathsf{T}}[k], \ldots, \tilde{\mathbf{w}}_{S,u}^{\mathsf{T}}[k]]^{\mathsf{T}}$ and $\mathbf{T}_u[k] \in \mathbb{C}^{N_{\mathrm{RF}} \times N_{\mathrm{RF}}}$. For all $i, j \in \mathcal{S}$, we have

$$\mathbf{T}_u^{i,j}[k] = \begin{cases} \bar{\alpha}_{i,u}^*[k]\, \bar{\alpha}_{j,u}[k]\, \mathbf{g}_{i,u}^* \mathbf{g}_{j,u}, & i \neq j, \\ \gamma_{i,u}[k] \mathbf{g}_{i,u}^* \mathbf{g}_{i,u}, & i = j, \end{cases} \tag{10}$$

where $\mathbf{T}_u^{i,j}[k] = [\mathbf{T}_u[k]]_{1+(i-1)N_{\mathrm{RF}}:iN_{\mathrm{RF}},\, 1+(j-1)N_{\mathrm{RF}}:jN_{\mathrm{RF}}}$. Note that the transition from (11a) to (11b) is obtained through straightforward algebraic manipulation, and the detailed steps are omitted for brevity. From the expression in (11b), we observe that the communication rate now depends solely on *statistical channel parameters* rather than instantaneous ones.

In the following, we formulate an optimization problem with respect to the digital beamformers (*hereafter referred to simply as beamformers*) across multiple LEO satellites, i.e., $\mathbf{W}_s[k], \forall s$, aiming to maximize the sum rate across all UTs. To reduce algorithmic complexity, we assume that the transmit power is uniformly allocated across subcarriers, allowing the beamformers to be designed independently for each subcarrier. This assumption aligns with practical protocols, where beamforming in OFDM systems is typically performed independently across different resource blocks [43]. *Accordingly, we omit the dependence on the subcarrier index k for notational simplicity.* The sum rate maximization problem, under per-satellite power budget, is then formulated as

$$\max_{\tilde{\mathbf{W}}_s} \quad \sum_{u=1}^{U} R_u^{\mathrm{LB}}$$
$$\text{s.t.} \quad (1-\delta_{s,u})\, \tilde{\mathbf{w}}_{s,u} = \mathbf{0}_{N_{\mathrm{RF}}},\ \forall s, u, \tag{12a}$$
$$\left\| \mathbf{F}_s \tilde{\mathbf{W}}_s \right\|_{\mathrm{F}}^2 \leq \frac{P_s}{K},\ \forall s, \tag{12b}$$

where $\tilde{\mathbf{W}}_s = [\tilde{\mathbf{w}}_{s,1}, \ldots, \tilde{\mathbf{w}}_{s,U}]$, and P_s denotes the power budget of the s-th LEO satellite. Here, (12a) ensures that $\tilde{\mathbf{w}}_{s,u} = \mathbf{0}_{N_{\mathrm{RF}}}$ whenever $\delta_{s,u} = 0$, i.e., the u-th UT is not be served by the s-th satellite.

B. WMMSE-Based Beamformer

In the following, we adopt the WMMSE optimization framework to reformulate the non-convex problem (12) into a sequence of convex quadratic problems by introducing auxiliary variables [12], [26]. This transformation enables efficient iterative optimization with guaranteed convergence to a stationary point [26]. Define the function

$$\Upsilon_u = \left| 1 - \mu_u \sum_{s=1}^{S} \bar{\alpha}_{s,u} \mathbf{g}_{s,u}^{\mathsf{T}} \tilde{\mathbf{w}}_{s,u} \right|^2 + |\mu_u|^2 \left(\underbrace{\sum_{s=1}^{S} \beta_{s,u} \left| \mathbf{g}_{s,u}^{\mathsf{T}} \tilde{\mathbf{w}}_{s,u} \right|^2 + \sum_{l \neq u}^{U} \tilde{\mathbf{w}}_u^{\mathsf{T}} \mathbf{T}_u \tilde{\mathbf{w}}_u + \sigma^2}_{\Psi_u} \right). \tag{13}$$

Here, μ_u and ν_u serve as the auxiliary variables [26]. With this reformulation, problem (12) becomes equivalent to the following

$$\max_{\mu_u, \nu_u, \tilde{\mathbf{W}}_s} \quad \sum_{u=1}^{U} (\ln \nu_u - \nu_u \Upsilon_u)$$
$$\text{s.t.} \quad (12a), (12b). \tag{14}$$

This problem can be efficiently solved through the alternating updates of $\tilde{\mathbf{W}}_s$, ν_u, and μ_u, as described below.

1) Update of μ_u: For fixed ν_u and $\tilde{\mathbf{W}}_s$, the optimal μ_u is obtained by minimizing Υ_u with respect to μ_u, i.e., by setting $\partial \Upsilon_u / \partial \mu_u = 0$. This yields the closed-form solution

$$\mu_u = \frac{\left(\sum_{s=1}^{S} \bar{\alpha}_{s,u} \mathbf{g}_{s,u}^{\mathsf{T}} \tilde{\mathbf{w}}_{s,u} \right)^*}{\left| \sum_{s=1}^{S} \bar{\alpha}_{s,u} \mathbf{g}_{s,u}^{\mathsf{T}} \tilde{\mathbf{w}}_{s,u} \right|^2 + \Psi_u}. \tag{15}$$

2) Update of ν_u: With μ_u and $\tilde{\mathbf{W}}_s$ fixed, the optimal ν_u that maximizes the objective in (14) is given by

$$\nu_u = \frac{1}{\Upsilon_u}. \tag{16}$$

3) Update of $\tilde{\mathbf{W}}_s$: Given the updated values of μ_u and ν_u, the optimal beamforming matrix $\tilde{\mathbf{W}}_s$ is found by solving the following optimization problem

$$\min_{\tilde{\mathbf{W}}_s} \quad \sum_{u=1}^{U} \nu_u \Upsilon_u$$
$$\text{s.t.} \quad (12a), (12b). \tag{17}$$

Here, Υ_u is convex quadratic function of $\tilde{\mathbf{W}}_s$. Therefore, (17) constitutes a convex quadratically constrained quadratic program (QCQP), which can be efficiently solved using convex optimization solvers such as CVX.

The iterative process for solving (12) is outlined in Algorithm 1. For completeness, we reintroduce the previously

$$R_u^{\mathrm{LB}}[k] = \log_2 \left(1 + \frac{\left| \mathbb{E}\left[\sum_{s=1}^{S} \alpha_{s,u}[k]\, \mathbf{g}_{s,u}^{\mathsf{T}} \tilde{\mathbf{w}}_{s,u}[k] \right] \right|^2}{\mathbb{V}\left[\sum_{s=1}^{S} \alpha_{s,u}[k]\, \mathbf{g}_{s,u}^{\mathsf{T}} \tilde{\mathbf{w}}_{s,u}[k] \right] + \sum_{l \neq u}^{U} \mathbb{E}\left[\left| \sum_{s=1}^{S} \alpha_{s,u}[k]\, \mathbf{g}_{s,u}^{\mathsf{T}} \tilde{\mathbf{w}}_{s,l}[k] \right|^2 \right] + \sigma^2} \right) \tag{11a}$$

$$= \log_2 \left(1 + \frac{\left| \sum_{s=1}^{S} \bar{\alpha}_{s,u}[k]\, \mathbf{g}_{s,u}^{\mathsf{T}} \tilde{\mathbf{w}}_{s,u}[k] \right|^2}{\sum_{s=1}^{S} \beta_{s,u}[k] \left| \mathbf{g}_{s,u}^{\mathsf{T}}[k]\, \tilde{\mathbf{w}}_{s,u}[k] \right|^2 + \sum_{l \neq u}^{U} \tilde{\mathbf{w}}_u^{\mathsf{H}}[k]\, \mathbf{T}_u[k]\, \tilde{\mathbf{w}}_u[k] + \sigma^2} \right) \tag{11b}$$

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Algorithm 1 Centralized Optimization for WMMSE-Based Networked LEO Satellite Distributed Beamforming

Initialize: $\tilde{\mathbf{W}}_s[k], \forall s, k$;
for $k = 1 : K$ **do**
  repeat
    Update $\mu_u[k]$ using (15);
    Update $\nu_u[k]$ using (16);
    Update $\tilde{\mathbf{W}}_s[k]$ by solving (17) via CVX;
  until the relative reduction in the objective value falls below a predefined threshold or a maximum number of iterations is reached;
end for
Output: $\tilde{\mathbf{W}}_s[k], \forall s, k$.

omitted subcarrier index k. The convergence of Algorithm 1 is established in [26]. The computational complexity of each iteration is primarily determined by solving (17), which, as a QCQP problem, is characterized by $\mathcal{O}((SN_{\text{RF}}T)^3)$.

Remark 3: Several important observations regarding computational complexity and scalability are worth highlighting:

- The analog beamformers are directly constructed via (9) without being included in the optimization loop. As a result, the computational complexity is significantly reduced from being dominated by the number of antennas N to being governed by the number of RFCs N_{RF}. This complexity reduction is particularly beneficial for future LEO satellite systems, which are expected to employ extremely massive antenna arrays [28].
- As indicated in (12), the digital beamformers at the LEO satellites, i.e., the $\tilde{\mathbf{W}}_s, \forall s$, are coupled within the objective function. Consequently, the WMMSE-based optimization framework necessitates a centralized implementation. In practice, a designated CPU among the LEO satellites collects the statistical CSI reported by all participating satellites and computes the intermediate parameters required for optimization, such as $\mathbf{v}_u$, $\mathbf{q}_u$, and $\mathbf{T}_u$. The beamformers are then jointly optimized at the CPU and subsequently distributed to the respective satellites. However, this centralized strategy imposes a heavy computational load on the CPU, leading to serious scalability concerns as the algorithmic complexity scales cubically with the number of satellites.

Remark 3 motivates us to optimize distributed beamforming in a decentralized manner by offloading computational tasks among the involved LEO satellites through the exploitation of ISLs, while *ensuring coherent computation toward optimizing a shared utility function*, such as the sum rate in this case. To this end, we briefly introduce ISL topologies below, which give rise to different decentralized beamforming designs.

C. ISL Topologies

ISLs serve as space infrastructure that enables satellites to communicate directly with each other via RF and/or free-space optical (FSO) signals, rather than relying on ground stations, which suffer from limited link budgets and high latency. ISLs offer high-speed and real-time communication capabilities for exchange of command, control, and data information [16], [44]. ISLs can be categorized into intra-plane and inter-plane types. Intra-plane ISLs allow satellites within the same orbital plane to communicate with each other. These links are relatively stable, as the satellites move at the same speed and maintain a consistent distance. In contrast, inter-plane ISLs enable communication between satellites in different orbital planes. These links are more dynamic because the relative distance and visibility between different orbits can vary significantly over time.

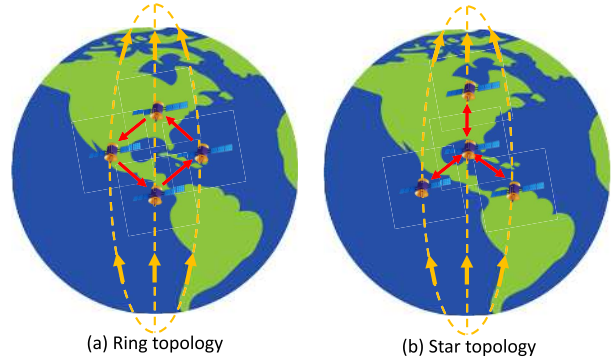

Fig. 2. Illustrations of different ISL topologies: (a) ring topology and (b) star topology.

As illustrated in Fig. 2, we present two representative topologies, namely, the Ring topology (Fig. 2(a)) and the Star topology (Fig. 2(b)), spanning across multiple orbital planes [45]. In the Ring topology, information flows unidirectionally through the network. In contrast, the Star topology enables bidirectional information exchange between edge satellites and a central satellite node. We implicitly assume that cooperation exists among LEO satellites in adjacent orbital planes, which traverse in the same direction [44]. This assumption facilitates the maintenance of stable ISLs topologies among cooperative LEO satellites and helps avoid frequent re-establishment of ISLs caused by changes in physical topology. In what follows, we develop two decentralized implementations of distributed beamforming tailored to networked LEO satellites systems, corresponding to the Ring and Star topologies, respectively.[4]

Note that practical ISL topologies often depart from the two idealized structures considered here and may involve more complex, time-varying configurations and constraints [45]. In addition, link-layer effects, such as synchronization jitter and queuing, introduce per-hop latency, and occasional packet losses may trigger retransmissions. These aspects primarily increase the elapsed time to convergence but do not fundamentally alter the optimization framework developed in the following sections. Thus, for simplicity, we abstract them away to focus on the algorithmic design. A comprehensive treatment, covering decentralized algorithms for *arbitrary* ISL topologies and practical mitigations (e.g., time-stamping and

[4]In this work, the ISL topology is modeled as *piecewise constant* within each beamforming-optimization window, whose duration is sub-second (typically tens to a few hundreds of ms). By contrast, handovers for a terrestrial UT at 500–600 km LEO altitudes occur on timescales of tens of seconds to a few minutes. This separation of timescales allows the distributed algorithms to converge under a fixed topology and only refresh when a handover or major topology change occurs. In practice, a handover triggers rescheduling with a warm start from the previous solution, which primarily increases convergence time rather than altering the optimizer or update rules.

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discarding stale messages, or reusing last-known intermediates when an update is missed) is left to future work.

IV. Distributed Beamforming Optimization Over Ring Topology-Based ISLs

In this section, we present a networked LEO satellite distributed beamforming scheme based on the Ring ISL topology. In this approach, the exchange of optimization intermediates among satellites is facilitated through a ring-structured data flow. We begin by revisiting (11), i.e., the lower bound of the ergodic rate, and approximating it in a manner amenable to decentralized computation. Then, we explain the decentralized WMMSE framework built upon the Ring topology-based ISLs.

A. Revisit of Hardening Bound of Ergodic Rate

The expression in (11b), though compact, is UT-centric and thus not suitable for satellite-centric decentralization. By omitting the dependence of the hardening bound on the subcarrier index for brevity, we rewrite (11a) as (18a), shown at the bottom of the page, where $\eta_{s,l} = \sum_{j\neq s}^{S}(\bar{\alpha}_{s,u}\mathbf{g}_{s,u}^{\mathsf{T}}\tilde{\mathbf{w}}_{s,l})^*(\bar{\alpha}_{j,u}\mathbf{g}_{j,u}^{\mathsf{T}}\tilde{\mathbf{w}}_{j,l})$, $\forall l \in \mathcal{S}/\{u\}$. ote that $\eta_{s,l}$ represents an IUI component that depends on coupled beamformers across multiple satellites. As a result, the contribution from any individual satellite cannot be separated without accounting for the others. This coupling complicates decentralized processing, as will be discussed later. Meanwhile, satellite communication typically operates in a noise-dominant regime due to severe signal attenuation over long propagation distances. In addition, as shown in Section II-D, the analog beamformer in (9) inherently suppresses IUI. Leveraging this property, we neglect the $\eta_{s,l}$ term in R_u^{LB} to derive the approximated ergodic rate in (18b), shown at the bottom of the page.

B. Decentralization of WMMSE Framework

After substituting the approximation given in (18b) into (12), we follow a reformulation similar to that presented in (13). The only difference is that the original expression of Υ_u in (13) is now replaced by

$$\Upsilon_u = |1-\mu_u F_u|^2 + |\mu_u|^2\left(P_u + \sum_{l\neq u}^{U} Q_{u,l} + \sigma^2\right), \quad (19)$$

where

$$F_u = \bar{\alpha}_{s,u}\mathbf{g}_{s,u}^{\mathsf{T}}\tilde{\mathbf{w}}_{s,u} + \underbrace{\sum_{j\neq s}^{S}\bar{\alpha}_{j,u}\mathbf{g}_{j,u}^{\mathsf{T}}\tilde{\mathbf{w}}_{j,u}}_{F_{s,u}}, \quad (20a)$$

$$P_u = \beta_{s,u}\left|\mathbf{g}_{s,u}^{\mathsf{T}}\tilde{\mathbf{w}}_{s,u}\right|^2 + \underbrace{\sum_{j\neq s}^{S}\beta_{j,u}\left|\mathbf{g}_{j,u}^{\mathsf{T}}\tilde{\mathbf{w}}_{j,u}\right|^2}_{P_{s,u}}, \quad (20b)$$

$$Q_{u,l} = \gamma_{s,u}\left|\mathbf{g}_{s,u}^{\mathsf{T}}\tilde{\mathbf{w}}_{s,l}\right|^2 + \underbrace{\sum_{j\neq s}^{S}\gamma_{j,u}\left|\mathbf{g}_{j,u}^{\mathsf{T}}\tilde{\mathbf{w}}_{j,l}\right|^2}_{Q_{s,u,l}}. \quad (20c)$$

We emphasize through (20) that the intermediate parameters F_u, P_u, and $Q_{u,l}$, which aggregate contributions from all satellites, can be decomposed into a component solely contributed by the s-th satellite and another aggregating the remaining $S-1$ satellites, i.e., $F_{s,u}$, $P_{s,u}$, and $Q_{s,u,l}$. At the s-th satellite, F_u, P_u, and $Q_{u,l}$ are made available through ISLs, we subsequently show that the auxiliary variables μ_u and ν_u, as well as the corresponding beamformer $\tilde{\mathbf{W}}_s$, can be updated *locally* (previously done at a CPU).

1) Local Update of μ_u and ν_u: Given F_u, P_u, and $Q_{u,l}$, by applying the update rules for auxiliary variables in the WMMSE framework, it is straightforward to derive the update equations for μ_u and $\nu_u$as

$$\mu_u = \frac{F_u^*}{|F_u|^2 + P_u + \sum_{l\neq u}^{U} Q_{u,l} + \sigma^2}, \quad \nu_u = \frac{1}{\Upsilon_u}. \quad (21)$$

2) Local Update of $\tilde{\mathbf{W}}_s$: With the updated values of μ_u and ν_u, similar to (17), the update of $\tilde{\mathbf{W}}_s$ should follow the principle of minimizing the objective function $\sum_{u=1}^{U}\nu_u\Upsilon_u$, which aggregates contributions from all satellites. To enable the local update of $\tilde{\mathbf{W}}_s$, we first reformulate the objective function (*by dropping the irrelevant terms regarding* $\tilde{\mathbf{W}}_s$) into

$$\sum_{u=1}^{U}\nu_u\Upsilon_u \Rightarrow -2\sum_{s=1}^{S}\Omega_s + \sum_{u=1}^{U}\nu_u|\mu_u|^2 Z_u, \quad (22)$$

where $\Omega_s = \sum_{u=1}^{U}\nu_u\Re\{\mu_u\bar{\alpha}_{s,u}\mathbf{g}_{s,u}^{\mathsf{T}}\tilde{\mathbf{w}}_{s,u}\}$ and $Z_u = |F_u|^2 + P_u + \sum_{l\neq u}^{U}Q_{u,l}$. Note that Ω_s denotes the component exclusively contributed by the s-th satellite. Here, the variables $F_{s,u}$, $P_{s,u}$, and $Q_{s,u,l}$, which are extracted from F_u, P_u, and $Q_{u,l}$, respectively, as defined in (20), are treated as *frozen* within Z_u, so that only the beamformer associated with the s-th satellite, i.e., $\tilde{\mathbf{W}}_s$ remains involved in the optimization.

It is worth mentioning that the values of $F_{s,u}$, $P_{s,u}$, and $Q_{s,u,l}$ must be computed prior to updating $\tilde{\mathbf{W}}_s$. These intermediate parameters are determined as

$$F_{s,u} = F_u - \bar{\alpha}_{s,u}\mathbf{g}_{s,u}^{\mathsf{T}}\tilde{\mathbf{w}}_{s,u}, \quad (23a)$$

$$P_{s,u} = P_u - \beta_{s,u}\left|\mathbf{g}_{s,u}^{\mathsf{T}}\tilde{\mathbf{w}}_{s,u}\right|^2, \quad (23b)$$

$$R_u^{\mathrm{LB}} = \log_2\left(1 + \frac{\left|\sum_{s=1}^{S}\bar{\alpha}_{s,u}\mathbf{g}_{s,u}^{\mathsf{T}}\tilde{\mathbf{w}}_{s,u}\right|^2}{\sum_{s=1}^{S}\beta_{s,u}\left|\mathbf{g}_{s,u}^{\mathsf{T}}\tilde{\mathbf{w}}_{s,u}\right|^2 + \sum_{l\neq u}^{U}\sum_{s=1}^{S}\left(\gamma_{s,u}\left|\mathbf{g}_{s,u}^{\mathsf{T}}\tilde{\mathbf{w}}_{s,l}\right|^2 + \eta_{s,l}\right) + \sigma^2}\right) \quad (18a)$$

$$\approx \log_2\left(1 + \frac{\left|\sum_{s=1}^{S}\bar{\alpha}_{s,u}\mathbf{g}_{s,u}^{\mathsf{T}}\tilde{\mathbf{w}}_{s,u}\right|^2}{\sum_{s=1}^{S}\beta_{s,u}\left|\mathbf{g}_{s,u}^{\mathsf{T}}\tilde{\mathbf{w}}_{s,u}\right|^2 + \sum_{l\neq u}^{U}\sum_{s=1}^{S}\gamma_{s,u}\left|\mathbf{g}_{s,u}^{\mathsf{T}}\tilde{\mathbf{w}}_{s,l}\right|^2 + \sigma^2}\right) \quad (18b)$$

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Algorithm 2 WMMSE-Based Networked LEO Satellite Distributed Beamforming Over a Ring Topology-Based ISLs

Initialize: $\tilde{\mathbf{W}}_s[k]$, $F_u[k]$, $P_u[k]$, $Q_{u,l}[k]$, $\forall s, u, l, k$;
for $k = 1 : K$ **do**
  repeat
    for $s = 1 : S$ **do**
      Receive $F_u[k]$, $P_u[k]$, and $Q_{u,l}[k]$ from the previous satellite;
      Update local variables $F_{s,u}[k]$, $P_{s,u}[k]$, and $Q_{s,u,l}[k]$ via (23);
      Update $\mu_u[k]$ and $\nu_u[k]$ using (21);
      Update $\tilde{\mathbf{W}}_s[k]$ by solving (24) via CVX;
      Update $F_u[k]$, $P_u[k]$, and $Q_{u,l}[k]$ via (20) and relay them to the next satellite;
    end for
  until the relative reduction in the objective value falls below a predefined threshold or a maximum number of iterations is reached;
end for
Output: $\tilde{\mathbf{W}}_s[k]$, $\forall s, k$.

$$Q_{s,u,l} = Q_{u,l} - \gamma_{s,u} \left| \mathbf{g}_{s,u}^{\mathsf{T}} \tilde{\mathbf{w}}_{s,l} \right|^2 . \tag{23c}$$

Given the updated μ_u and ν_u, the local subproblem for optimizing $\tilde{\mathbf{W}}_s$ can be formulated as

$$\min_{\tilde{\mathbf{W}}_s} \quad -2\Omega_s + \sum_{u=1}^{U} \nu_u \left|\mu_u\right|^2 Z_u$$
$$\text{s.t.} \quad (1-\delta_{s,u})\, \tilde{\mathbf{w}}_{s,u} = \mathbf{0}_{N_{\text{RF}}},\ \forall u, \tag{24a}$$
$$\left\| \mathbf{F}_s \tilde{\mathbf{W}}_s \right\|_{\text{F}}^2 \leq \frac{P_s}{K}. \tag{24b}$$

It is straightforward to see that (24) is a convex QCQP, which can be efficiently solved using CVX.

C. Overall Workflow

Based on the above discussion, the decentralized deployment of the WMMSE-based algorithm for networked LEO satellite distributed beamforming over the Ring topology-based ISLs can be summarized as follows. Without loss of generality, the intermediate parameters F_u, P_u, and $Q_{u,l}$ are relayed from the $(s-1)$-th satellite to the s-th satellite. After reaching the S-th satellite, these variables are forwarded back to the first satellite, completing the ring. At each satellite, as illustrated in Fig. 3(a), once the intermediate parameters F_u, P_u, and $Q_{u,l}$ are received, the variables $F_{s,u}$, $P_{s,u}$, and $Q_{s,u,l}$ are first updated according to (23), before the previous beamformer $\tilde{\mathbf{W}}_s$ is overwritten. Then, the auxiliary variables μ_u and ν_u, as well as the beamformer $\tilde{\mathbf{W}}_s$, are sequentially updated according to (21) and (24). Subsequently, the updated F_u, P_u, and $Q_{u,l}$ are computed using (20) and relayed to the next satellite in the ring dataflow.

The entire process, following the redemption of the subcarrier index k, is summarized in Algorithm 2. The convergence of Algorithm 2 directly follows from the use of the WMMSE framework. As a decentralized algorithm, the per-satellite computational complexity in each iteration is dominated by solving (24), which, due to its QCQP nature, has a complexity of $\mathcal{O}((N_{\text{RF}}T)^3)$. Note that the use of steering-vector-based analog beamforming reduces the optimization dimension from the order of the number of antennas to that of the RFCs, which are typically much fewer. This leads to a substantial complexity reduction, making the proposed design more practical and implementation-friendly for resource-constrained LEO satellites.

Remark 4: We conclude that the algorithmic complexity is offloaded from the CPU to the participating LEO satellites by the proposed distributed beamforming scheme in Algorithm 2, significantly enhancing the scalability of satellite cooperation. However, despite the advantage in scalability, one potential drawback of Algorithm 2 is that the decentralized deployment incurs additional ISL backhaul overhead for relaying the intermediate parameters $F_u[k]$, $P_u[k]$, and $Q_{u,l}[k]$ ($\forall s, u, l, k$), compared to the centralized deployment in Algorithm 1. The total dimensionality of this per-iteration ISL backhaul overhead is given by $D_{\text{Ring}} = K(U^2 + 2U)$. Note also that due to the iterative nature of the WMMSE-based solution, the number of iterations required for convergence may result in longer latency compared to the centralized deployment, owing to the propagation delay associated with relaying intermediate parameters via ISLs.

V. Distributed Beamforming Optimization Over Star Topology-Based ISLs

In this section, we propose a networked LEO satellite distributed beamforming approach by adapting the WMMSE optimization framework to the Star topology-based ISLs. The exchange of intermediate optimization parameters among the involved satellites is facilitated by a star-structured workflow. We begin by highlighting the unique requirement of distributed beamforming optimization via Star topology-based ISLs, namely, achieving *consensus*, followed by a recap of the overall protocol.

A. Achieving Consensus

In contrast to the Ring topology where each involved satellite plays a symmetric role and local processing proceeds unidirectionally through the network, the Star topology features a central satellite that exchanges information bidirectionally with the edge satellites (see Fig. 3). Thus, in the distributed beamforming framework based on the Star topology, the edge satellites, upon receiving the initial intermediate parameters F_u, P_u, and $Q_{u,l}$ from the central satellite, simultaneously carry out local updates of the variables $F_{s,u}$, $P_{s,u}$, and $Q_{s,u,l}$, the auxiliary variables μ_u and ν_u, the beamformer $\tilde{\mathbf{W}}_s$, and the intermediate parameters F_u, P_u, and $Q_{u,l}$, instead of performing local update sequentially. Note that this local update process is also performed at the central satellite, concurrently with the updates at the edge satellites. Afterwards, F_u, P_u, and $Q_{u,l}$ updated at the edge satellites are fed back to the central satellite. Since F_u, P_u, and $Q_{u,l}$ are updated in parallel by different edge satellites as well as the central satellite, discrepancies may arise in their values. Therefore, consensus over these intermediate parameters must be achieved at the central satellite before proceeding to the next iteration.

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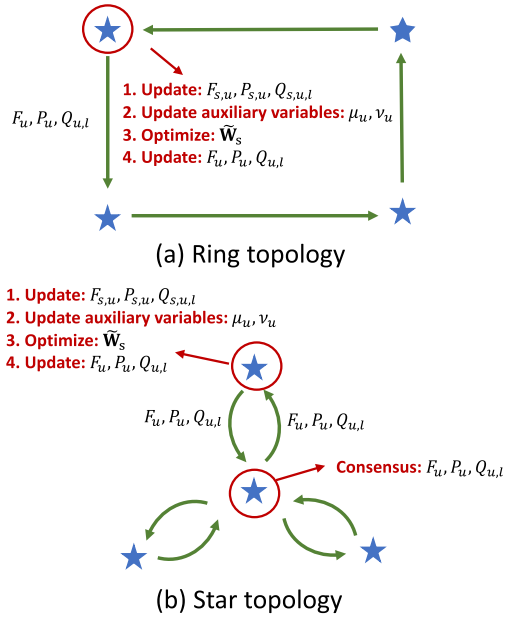

Fig. 3. Workflow under different ISL topologies: (a) ring topology and (b) star topology.

Let $\mathbf{f} = [F_1, \ldots, F_U]^\mathsf{T}$, $\mathbf{p} = [P_1, \ldots, P_U]^\mathsf{T}$, and $\mathbf{Q} \in \mathbb{R}^{U\times U}$ be the matrix whose (u,l)-th element is $Q_{u,l}$. Note that, to maintain generality, we do not explicitly specify which satellite serves as the central satellite in the network. To ensure consistency, the intermediate parameters must reach consensus under a unified objective within the overall optimization framework, i.e., maximizing $\sum_{u=1}^{U}(\ln \nu_u - \nu_u \Upsilon_u)$, which is equivalent to minimizing $\sum_{u=1}^{U} \nu_u \Upsilon_u$ for a given ν_u. Recall that Υ_u is rewritten as a function of $\mathbf{f}$, $\mathbf{p}$, and $\mathbf{Q}$ in (19). The corresponding optimization problem to determine $\mathbf{f}$, $\mathbf{p}$, and $\mathbf{Q}$, while achieving consensus, is formulated as

$$\min_{\mathbf{f},\mathbf{p},\mathbf{Q}} \quad \sum_{u=1}^{U} \nu_u \Upsilon_u$$
$$\text{s.t.} \quad \mathbf{f} = \mathbf{f}_s, \quad \mathbf{p} = \mathbf{p}_s, \quad \mathbf{Q} = \mathbf{Q}_s, \quad \forall s, \tag{25a}$$
$$\mathbf{p} \geq \mathbf{0}_U, \quad \text{vec}\left(\mathbf{Q}\right) \geq \mathbf{0}_{U^2}, \tag{25b}$$

where $\mathbf{f}_s$, $\mathbf{p}_s$, and $\mathbf{Q}_s$ are parameters updated by the s-th satellite, and constraint (25b) is inherently required by the definitions in (20b) and (20c).

The above convex problem with equality constraints can be efficiently solved using the PDD framework [46], [47], which relaxes complex constraints by adding them as penalty terms at the objective function. It iteratively updates both primal and dual variables to achieve balance between feasibility and optimality while keeping the problem tractable. Specifically, a double-loop structure is employed: The inner loop optimizes the augmented Lagrangian problem (ALP) of the original formulation, while the outer loop updates the Lagrangian dual variables and penalty parameters [46].

For the inner loop of the PDD framework, the ALP corresponding to (25) is formulated as

$$\min_{\mathbf{f},\mathbf{p},\mathbf{Q}} \quad \sum_{u=1}^{U} \nu_u \Upsilon_u + \frac{\rho}{2} \sum_{s=1}^{S} \left\| \mathbf{\Gamma} - \mathbf{\Gamma}_s + \frac{1}{\rho} \mathbf{\Theta}_s \right\|_{\mathrm{F}}^2$$
$$\text{s.t.} \quad \mathbf{p} \geq \mathbf{0}_U, \quad \text{vec}\left(\mathbf{Q}\right) \geq \mathbf{0}_{U^2}, \tag{26}$$

Algorithm 3 PDD-Based Consensus Algorithm

Initialize: $\mathbf{\Theta}_s = \mathbf{0}_{(U+2)\times U}, \forall s$, ρ, δ, $j = 1$;
repeat
  Update $(\mathbf{f}, \mathbf{p}, \mathbf{Q})$ by solving (26);
  if $h\left(\mathbf{\Gamma}\right) \leq \zeta^{(j)}$ **then**
    $\mathbf{\Theta}_s = \mathbf{\Theta}_s + \rho\left(\mathbf{\Gamma} - \mathbf{\Gamma}_s\right), \forall s$;
  else
    $\rho = \delta\rho$;
  end if
  $j = j + 1$
until $h\left(\mathbf{\Gamma}\right)$ is below a specified threshold or a maximum number of iterations is reached;
Output: $\mathbf{f}$, $\mathbf{p}$, $\mathbf{Q}$.

where $\mathbf{\Gamma} = [\mathbf{f}, \mathbf{p}, \mathbf{Q}^\mathsf{T}]^\mathsf{T}$ and $\mathbf{\Gamma}_s = [\mathbf{f}_s, \mathbf{p}_s, \mathbf{Q}_s^\mathsf{T}]^\mathsf{T}$. Here, $\mathbf{\Theta}_s \in \mathbb{C}^{(U+2)\times U}$ and ρ denote the Lagrangian dual variable and penalty parameter, respectively. Note that the inner ALP problem in (26) is a convex QCQP, and thus its optimal solution can be efficiently obtained using CVX directly, without the need for an inner loop by block coordinate descent [46].

Following the principle of the PDD framework [46], to ensure convergence, we define the violation function as $h(\mathbf{\Gamma}) = \min\left(h_1(\mathbf{\Gamma}), \ldots, h_S(\mathbf{\Gamma})\right)$, where $h_s(\mathbf{\Gamma}) = \|\mathbf{\Gamma} - \mathbf{\Gamma}_s + \frac{1}{\rho}\mathbf{\Theta}_s\|_\infty$. The detailed steps for updating the Lagrangian dual variable $\mathbf{\Theta}_s$ and the penalty parameter ρ in the outer loop, along with the overall procedure for solving (25) using the PDD approach, are summarized in Algorithm 3. Here, $\delta > 1$ is a constant used to increase the penalty parameter when necessary, while $\zeta^{(j)}$ denotes an empirically defined sequence that asymptotically approaches zero. Specifically, we define $\zeta^{(j)} = qh^{(j-1)}(\mathbf{\Gamma})$, where $q \in (0,1)$ is an attenuation constant, and $h^{(j-1)}(\mathbf{\Gamma})$ is the value of $h(\mathbf{\Gamma})$ at the $(j-1)$-th iteration.

B. Overall Workflow

Built upon the above, the decentralized deployment process of the WMMSE-based algorithm for networked LEO satellite distributed beamforming over a Star topology-based ISLs, following the redemption of the subcarrier index k, is summarized in Algorithm 4. Similar to Algorithm 2, the per-satellite computational complexity for the local update in each iteration is dominated by solving (24), which has a complexity of $\mathcal{O}((N_{\mathrm{RF}}T)^3)$. At the central satellite, the additional requirement of achieving consensus introduces further complexity, primarily from solving (26) in each iteration, with a complexity of $\mathcal{O}((U(U+2))^3)$.

Remark 5: Steps 4–11 of Algorithm 4, corresponding to the local beamformer updates at all satellites, are executed in parallel. In contrast, the local updates in the Ring topology are carried out sequentially. As will be shown in the simulations, this parallelism reduces the number of iterations required for convergence, potentially resulting in lower latency, especially when accounting for the propagation delay associated with intermediate parameter exchanges over ISLs. The total dimensionality of the per-iteration ISL communication overhead from the perspective of an edge satellite is the same as that under the Ring topology, denoted by $D_{\mathrm{Star}}^{\mathrm{E}} = K(U^2 + 2U)$. However, due to parallelism, the per-iteration ISL

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Algorithm 4 WMMSE-Based Networked LEO Satellite Distributed Beamforming Over a Star Topology-Based ISLs

Initialize: $\mathbf{W}_s[k]$, $F_u[k]$, $P_u[k]$, $Q_{u,l}[k]$, $\forall s, u, l, k$;
for $k = 1 : K$ **do**
  repeat
    for $s = 1 : S$ **do**
      Receive $F_u[k]$, $P_u[k]$, and $Q_{u,l}[k]$ from the central satellite;
      Update $\mu_u[k]$ and $\nu_u[k]$ using (21);
      Update local variables $F_{s,u}[k]$, $P_{s,u}[k]$, and $Q_{s,u,l}[k]$ via (23);
      Update $\mathbf{W}_s[k]$ by solving (24) via CVX;
      Update $F_u[k]$, $P_u[k]$, and $Q_{u,l}[k]$ via (20) and relay them to the central satellite;
    end for
    At the central satellite, achieve consensus over $F_u[k]$, $P_u[k]$, and $Q_{u,l}[k]$ via Algorithm 3;
    Send the consensused $F_u[k]$, $P_u[k]$, and $Q_{u,l}[k]$ to all edge satellites;
  until the relative reduction in the objective value falls below a predefined threshold or a maximum number of iterations is reached;
end for
Output: $\mathbf{W}_s[k]$, $\forall s, k$.

communication overhead from the perspective of the central satellite is denoted by $D^{\mathrm{C}}_{\mathrm{Star}} = K(S-1)(U^2+2U)$.

VI. Numerical Results

A. Simulation Setting

We model the Earth as a sphere with a radius of 6400 km. The region of interest, where UTs are randomly distributed, lies within a circular service area on the sphere with a radius of 200 km [21]. The participating LEO satellites are distributed within a parallel circular area, whose center is located along the extension of the line from the Earth's core to the center of the service area, at an orbital height of 500 km. The UPA installed on each satellite is oriented tangentially to the orbit, with its local coordinate system's z-axis pointing toward the Earth's core and its x-axis perpendicular to the orbital plane [12]. The components that constitute the large-scale path loss $\beta_{s,u}$ are generated according to (5), as well as the elaborations and remarks that follow. The Rician factors are randomly chosen between 15 and 20 dB [21]. The remaining simulation parameters are listed in Table III.

TABLE III
Simulation Parameters

Parameter	Value
Carrier frequency f_c	12.7 GHz (Ku band)
Subcarrier spacing Δf	120 KHz
Subcarrier number K	1024
Power budget at each LEO satellite	50 dBm
PSD N_0	-173.855 dBm/Hz
Noise figure F	10 dB
Number of LEO satellites S	4
Number of UTs U	16
Number of RFC N_{RF}	8
Antenna number $N = N_{\mathrm{h}} \times N_{\mathrm{v}}$	16×16
Antenna radiation gain $G(\theta)$	$\sqrt{\frac{3}{4\pi}}\cos(\theta)$ [27]

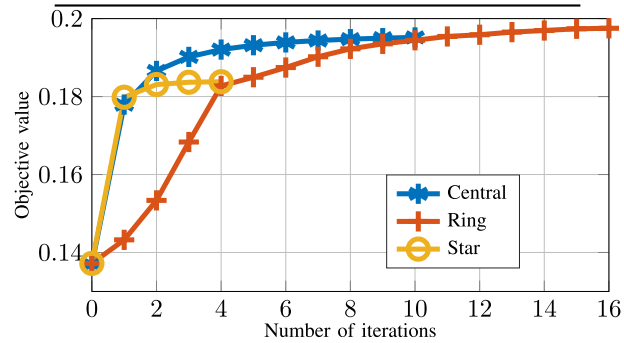

Fig. 4. Convergent behavior of the proposed networked LEO satellite distributed beamforming schemes.

B. Benchmark Schemes

We evaluate the performance of three proposed WMMSE-based distributed beamforming schemes for networked LEO satellites: one centralized scheme and two decentralized schemes operating under Ring and Star topologies of ISLs, referred to as Central, Ring, and Star for brevity. For comparison, we introduce two networked LEO satellite beamforming baselines (under the same user scheduling policy):

- MRT: At each LEO satellite, the beamformer is denoted by $\zeta_s \mathbf{w}_{s,u}[k]$. If $u \in \mathcal{U}_s$, $\mathbf{w}_{s,u}[k]$ is designed to align with $\mathbf{g}_{s,u}$; otherwise, $\mathbf{w}_{s,u}[k] = \mathbf{0}_T$. Here, ζ_s is a normalization factor that ensures compliance with the power constraint. This method is also used to initialize the beamformers in the proposed centralized and decentralized WMMSE-based approaches;
- ZF: At each LEO satellite, if $u \in \mathcal{U}_s$, the beamformer is designed to lie in the null space of $\{\mathbf{g}_{s,j}\}_{j \in \mathcal{U}_s \setminus \{u\}}$, with a normalization factor for power constraint. Otherwise, $\mathbf{w}_{s,u}[k] = \mathbf{0}_T$.

To further highlight the advantages of cooperative multi-satellite systems, we also consider three baselines based on single-satellite service (S^3), where each UT is exclusively served by the LEO satellite closest to it. Under the S^3 setting, we apply the WMMSE, MRT, and ZF beamforming individually, denoted as WMMSE-S^3, MRT-S^3, and ZF-S^3, respectively. Specifically, for WMMSE-S^3, the beamformer optimization is carried out independently at each satellite without considering potential inter-satellite interference.

C. Simulation Results

In Fig. 4, we investigate the convergence behavior of the proposed WMMSE-based networked LEO satellite distributed beamforming approaches. We observe that, compared to Central, Ring converges to a comparable objective value, albeit at the cost of requiring more iterations. In contrast, Star converges to a lower objective value with fewer iterations, owing to its structure where edge satellites update in parallel without waiting for results from others. It is important to highlight that, unlike Central where iterations are executed internally within the CPU, iterations of Ring and Star involve the exchange of intermediate parameters via ISLs, introducing additional latency (see Remarks 4 and 5). Consequently, although Ring

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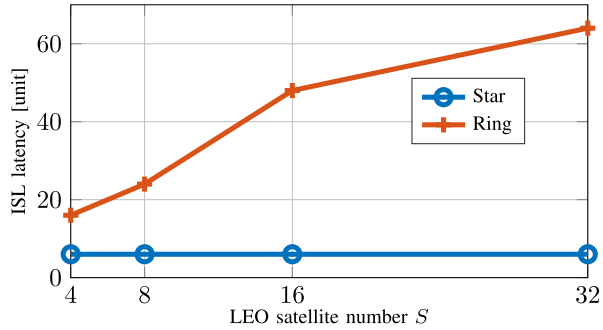

Fig. 5. ISL latency versus LEO satellite number S.

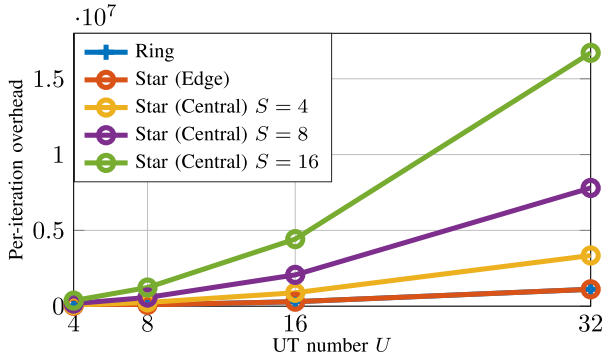

Fig. 6. Per-iteration communication overhead versus UT number U.

eventually converges to a higher value than Star, the excessive number of iterations could compromise its effectiveness due to practical coherence degradation over multiple iterations.

In Fig. 5, we compare the incurred ISL latency as a function of the number of LEO satellites between Ring and Star. *Note that, for simplicity, we ignore potential propagation latency differences across different ISLs, and assume that each ISL incurs one unit of latency.* Additionally, for Ring, each algorithmic iteration incurs one unit of ISL latency (from one satellite to its immediate neighbor), whereas for Star, each iteration incurs two units of ISL latency (from the central node to an edge node and back). As observed, the total ISL latency for the Star topology remains constant as the number of participating satellites increases, while it grows linearly for the Ring topology. This behavior arises from the fundamentally different decentralization mechanisms employed by the two schemes. In the Star topology, the inherent parallelism among the edge satellites ensures that scaling the network with more satellites does not significantly increase the overall latency. In contrast, in the Ring topology, the sequential nature of local updates means that involving more satellites naturally leads to higher latency, as the total one-loop latency is proportional to the number of satellites.

In Fig. 6, we compare the per-iteration communication overhead in terms of the overall dimension of the conveyed intermediate parameters versus the number of UTs for the Ring and Star topologies. As discussed in Remark 5, unlike the symmetric unidirectional information exchange in Ring, Star requires asymmetric bidirectional communication. From the perspective of edge satellites, the communication overhead in Star is comparable to that in Ring. However, for the central satellite in Star, the overhead scales by a factor of $S - 1$. As shown, all overheads increase with the number

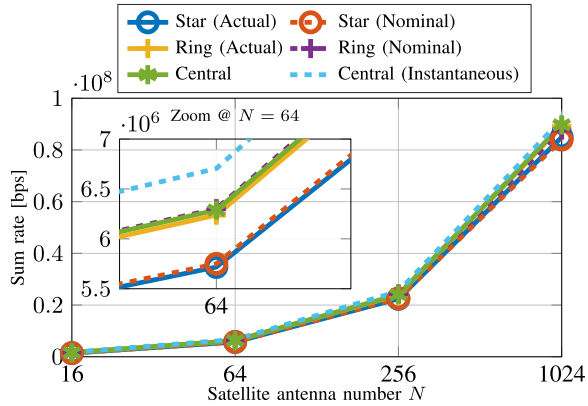

Fig. 7. Sum rate performance versus antenna number N, comparing the proposed statistical CSI-based beamforming optimization (both nominal and actual) with the average sum rate achieved under instantaneous CSI-based beamforming optimization.

of participating UTs. Notably, while the overhead for Ring and for the edge satellites in Star remains independent of the number of satellites, the overhead for the central satellite in Star grows as more satellites participate.

The key insight from the above analysis is that, although the Star topology attains a substantial fraction of the centralized gain within a modest number of iterations, its signaling overhead at the central satellite scales unfavorably with the numbers of satellites and UTs. In contrast, the Ring topology eliminates the central bottleneck and can achieve a higher rate upon convergence, but it typically requires many more iterations, resulting in larger latency. These observations suggest moving beyond pure Ring or Star toward more flexible architectures. Promising directions include (i) hybrid topologies (e.g., Ring–Star) that trade central overhead for latency, (ii) fully decentralized consensus that does not rely on a central node, and (iii) exchange of dimension-compressed intermediate parameters to reduce signaling. A full exploration is beyond the scope of this paper and is left to future work.

Fig. 7 illustrates the sum rate performance as a function of the satellite antenna number N, comparing the actual and nominal rates under the proposed Ring- and Star-based decentralized optimization algorithms, thereby quantitatively validating the approximation in (18b). The nominal rate is obtained from the approximated expression in (18b), where the $\eta_{s,l}$ term representing part of the cross-satellite interference is omitted, while the actual rate is calculated from the complete expression in (18a). For reference, the centralized optimization result is also included as an upper bound. The results show that the difference between the actual and nominal rates remains marginal across the entire range of antenna numbers, confirming the validity of the approximation. This outcome is consistent with the discussion in Section IV-A: satellite communication generally operates in a noise-dominant regime due to long propagation distances, while the analog beamformer in (9) inherently suppresses interference. As a result, omitting the $\eta_{s,l}$ term, which represents part of the interference component, leads to negligible performance variation.

In addition, Fig. 7 also evaluates the performance gap between the proposed statistical CSI-based framework and schemes based on instantaneous CSI. Specifically, the figure

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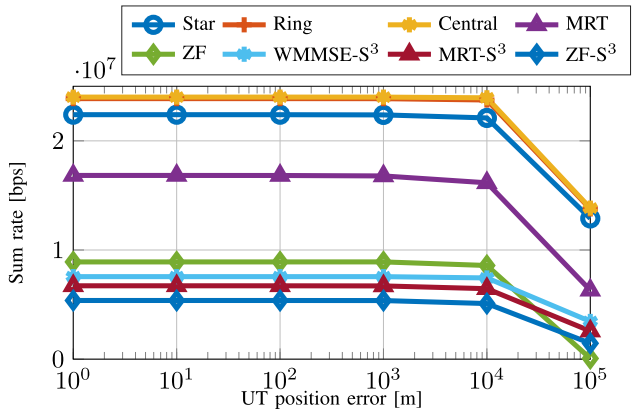

Fig. 8. Sum rate under various schemes versus UT position error.

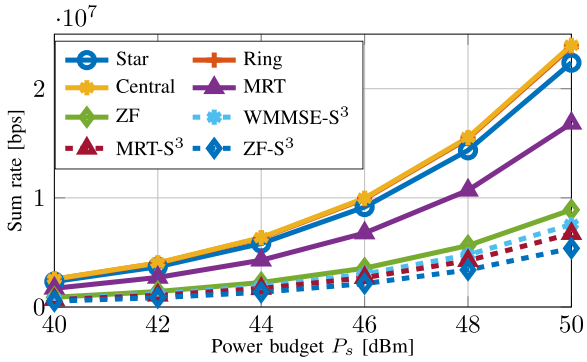

Fig. 9. Sum rate under various schemes versus power budget P_s.

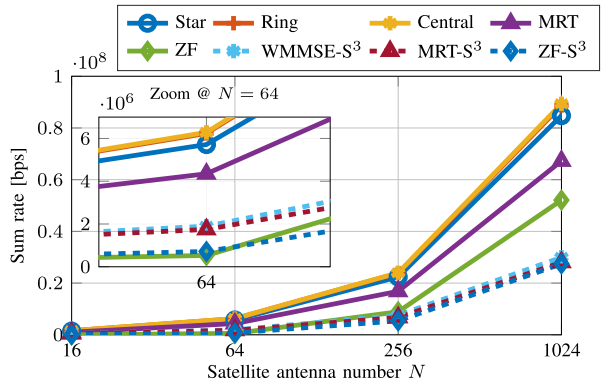

Fig. 10. Sum rate under various schemes versus antenna number N.

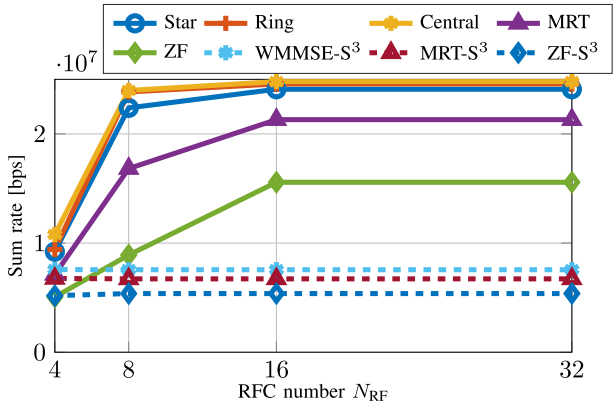

Fig. 11. Sum rate under various schemes versus the number RFCs N_{RF}.

presents the sum rate performance of instantaneous CSI-based distributed beamforming optimized using the WMMSE framework, with results averaged over 100 Monte Carlo trials to ensure statistical reliability. As observed, the performance loss of the proposed statistical CSI-based optimization is marginal compared to the instantaneous CSI-based benchmark. This robustness is attributed to the LOS-dominant nature of LEO satellite channels, where the steering-vector-based analog beamforming stage effectively suppresses interference prior to digital optimization, thereby reducing the relative advantage of instantaneous CSI for interference management.

Fig. 8 shows the sum rate performance as a function of UT position error, evaluating the robustness of the proposed analog beamforming design under imperfect AOD estimates. The results compare the proposed schemes with benchmark designs and indicate that noticeable degradation occurs only when the position error reaches the order of 10 km. Since modern GNSS systems typically provide meter-level accuracy (except in rare cases of severe outage), such large errors are highly unlikely in practice. This robustness stems from the fact that UT–satellite distances are on the order of 500 km, so moderate position errors translate into negligible angular mismatches. Overall, the figure confirms that the proposed analog beamforming approach remains resilient under realistic conditions with imperfect position information.

In Figs. 9 and 10, we compare the sum rate of various schemes as functions of the power budget and the number of antennas, respectively. It is observed that among the networked LEO satellite distributed beamforming approaches, the proposed WMMSE-based methods achieve substantially higher sum rate compared to the heuristic strategies (i.e., MRT and ZF), demonstrating the effectiveness of the proposed optimization schemes. Within the WMMSE-based designs, Ring yields similar performance compared to Central, while Star is slightly outperformed by Central. Moreover, the networked LEO distributed beamforming schemes significantly outperform the baseline approach, where each UT is individually served by a single satellite (i.e., the schemes with the suffix S^3). This performance gap widens as the power budget and antenna number increase, highlighting the substantial benefits of cooperative satellite transmission in enhancing communication efficiency.

In Fig. 11, we plot the sum rate under various numbers of RFCs and various schemes. The proposed WMMSE-based distributed beamforming approaches consistently outperform the baselines, with the sum rate steadily increasing as the number of RFCs grows. Nevertheless, the growth of the sum rate saturates beyond a certain RFC threshold, reflecting the saturation of the available digital-domain degrees-of-freedoms (DOFs) enabled by RFC deployment. Across the entire comparison range, the performance gap between the decentralized Ring and Star topologies and their centralized counterpart (i.e., Central) remains mild, highlighting the robustness of the proposed decentralized networked LEO satellite distributed beamforming schemes. It is also noteworthy that increasing the number of RFCs does not substantially enhance the sum rate under the S^3 approaches, as fewer UTs are scheduled to each satellite (due to the non-overlapping user scheduling), causing RFC numbers to reach abundance more quickly.

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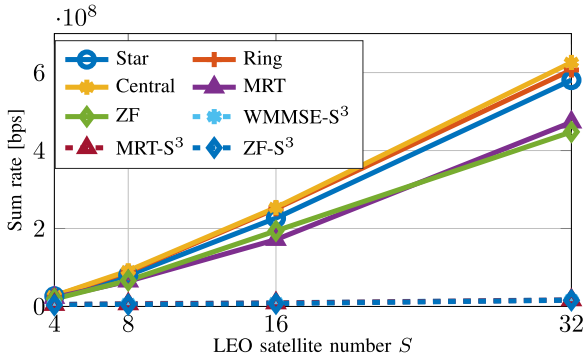

Fig. 12. Sum rate under various schemes versus LEO satellite number S.

In Fig. 12, we plot the sum rate as a function of the number of participating LEO satellites under various schemes. As observed, the sum rate increases significantly with the number of satellites involved in all networked LEO satellite distributed beamforming schemes, whereas it remains nearly constant under the S^3 approaches. This is because the networked LEO satellite schemes leverage the enlarged equivalent array aperture and enhanced cooperative array gain as more satellites participate. In contrast, under the S^3 approaches, each UT is served individually by a single satellite, with satellites operating independently. This limits the ability to fully exploit the collective capabilities of the satellite constellation and prevents the improvement of spectral efficiency through cooperative transmission. These results further highlight the substantial advantages of networked LEO satellite distributed beamforming. Moreover, it is evident that the performance gap between the WMMSE-based approaches and other heuristic networked LEO satellite schemes (MRT and ZF) widens as the number of participating satellites increases, demonstrating the effectiveness of the proposed optimization method.

VII. Conclusion

This paper addressed a critical gap in the design of distributed beamforming schemes for networked LEO satellite systems by developing scalable solutions that operate solely on statistical CSI. We established a comprehensive system model for networked LEO satellite OFDM downlink communications, incorporating large antenna arrays with limited RFCs, and introduced practical yet effective designs for user scheduling and analog beamforming. A closed-form lower bound on the ergodic sum rate was derived using the hardening bound, forming the basis for a sum rate maximization problem solved centrally via the WMMSE framework. Building upon this, we proposed two decentralized optimization schemes tailored to Ring and Star ISLs topologies, enabling local beamformer updates and cooperative parameter exchange with distinct information flows. Extensive simulations demonstrated that the proposed distributed methods closely approached the performance of centralized method, significantly outperformed baseline schemes, and offered clear scalability advantages. Furthermore, the observed trade-offs between delay and communication overhead across topologies provided valuable insights for future research in large-scale, dynamic LEO satellite cooperation. An important direction for future work is the development of lightweight decentralized algorithms with fewer iteration requirements and adaptive mechanisms that can explicitly address satellite mobility, topology variations, and handovers, thereby enhancing the practicality of distributed optimization in dynamic LEO networks.

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